Submission 2276: A Dual-Expert Strategy Integrating LLMs to Mitigate Negative Transfer in Cross-Domain Sequential Recommendation

A Baselines

In this section, we review key previous studies that have addressed the CDSR task (baselines compared in Tables 2, 3, 7, and 8).

A.1 IDRec for CDSR

CGRec [23] and SyNCRec [24] represent pioneering studies in Cross-Domain Sequential Recommendation (CDSR) as all-in-one models capable of learning from three or more domains simultaneously. In contrast, methods such as MAN [14], C2DSR [1], π-Net and MIFN [20] are specifically designed to address CDSR challenges within two-domain pairs, limiting their applicability to multi-domain scenarios. For this reason, we present a detailed performance result of these models, which are limited to handling two-domain pairs, separately in Table 7 and 8 , as opposed to the main performance results discussed in Table 2 and 3.

- **CGRec** [23] addresses the challenge of negative transfer in CDSR by proposing a framework that adaptively reduces the influence of dissimilar domains using cooperative game theory to assess domain contributions. Additionally, a hierarchical contrastive learning approach leveraging coarse- and fine-level category information enhances model performance, achieving state-of-the-art results on real-world datasets.
- **SyNCRec** [24] also tackles negative transfer in CDSR by introducing an adaptive weighting mechanism based on the degree of negative transfer and leveraging mutual information between CDSR and SDSR tasks.
- **MoSE** [25] addresses the challenge of modeling sequential user behavior in multi-task learning settings, incorporating Long Short-Term Memory (LSTM) into the Multi-gate Mixture-of-Experts architecture. It effectively captures temporal dependencies and handles heterogeneous data sources like web search and browsing logs with differing properties.
- **MAN** [14] addresses data sparsity in sequential recommendation by leveraging cross-domain information without relying on overlapped users. It employs local and global attention modules to capture domain-specific and domain-general patterns, utilizing a mixed attention layer and prediction layer to effectively fuse and evolve user interests, demonstrating superior performance on real-world datasets.
- **C^{2}DSR** [1] simultaneously captured single- and cross-domain user preferences by utilizing a graph neural network and a sequential attentive encoder to model intra- and inter-sequence item relationships. It further incorporated a contrastive cross-domain infomax objective to strengthen the connection between single- and cross-domain user representations.
- **π-Net** [21] addresses the Shared-account Cross-domain Sequential Recommendation (SCSR) task by introducing a Parallel Information-sharing Network that simultaneously generates recommendations for two domains. It employs a Cross-Domain Transfer Unit (CTU) to adaptively transfer information across domains, achieving improved recommendations by integrating cross-domain user behavior.
- **MIFN** [20] enhances the performance of CDSR task by modeling both the flow of behavioral information and knowledge across domains using behavior and knowledge transfer units. This mixed information flow network selectively utilizes cross-domain information based on users' current preferences, achieving improved performance on e-commerce datasets.

A.2 LLMRec

LLMRec is a rapidly emerging field of research that integrates the semantic features of user behavior sequences, represented in textual form, with the open-world knowledge of Large Language Models (LLMs). As a result, LLMRec demonstrates competitive performance even in cross-domain settings.

- **UniSRec** [6] utilizes pre-trained language models to encode item texts and introduces a parametric whitening and mixture-of-experts (MoE) adaptors to create isotropic, transferable item representations for sequential recommendation.
- **E4SRec** [12] focuses on a lightweight integration of LLMs with traditional ID-based recommendation systems, using minimal pluggable components to enhance efficiency and scalability.
- **RecFormer** [11] introduces a framework for ID-free sequential recommendation by learning language representations through a bi-directional Transformer model that encodes item key-value attributes as text. This approach improves transferability across domains and cold-start settings, demonstrating significant performance gains in both supervised and zero-shot recommendations.
- **SAID** [8] proposes a two-step approach to sequential recommendation. In the first step, a projector module maps item IDs into embeddings that are semantically compatible with the representation space of large language models (LLMs). In the second step, these aligned embeddings are fed into efficient sequential models—such as GRU or Transformer—that are trained to perform next-item prediction.
- **CALRec** [13] introduces a two-phase fine-tuning strategy for generative LLMs, combining joint training across multiple categories with specialized tuning for individual categories. The model is trained using a hybrid loss that merges next-item generation with contrastive alignment objectives, and it leverages a quasi-round-robin BM25-based retrieval method to enhance recommendation efficiency. We performed inference using cosine similarity–based matching in our practical implementation.
- **ReLLa** [15] introduces a framework that enhances Large Language Models (LLMs) for recommendation tasks by addressing the challenge of long user behavior sequences using two key components: semantic user behavior retrieval (SUBR) and retrieval-enhanced instruction tuning (ReiT). SUBR improves data quality by selecting semantically relevant behaviors, while ReiT augments training samples with retrieval-enhanced counterparts to increase robustness and

generalization.This design enables ReLLa to outperform traditional models even in few-shot settings, leveraging LLMs' reasoning abilities to comprehend lifelong sequential behavior, making it effective for both zero-shot and few-shot recommendation scenarios.

- **Lite-LLMRec** [30] proposes a hierarchical LLM structure with two components: an Item LLM to encode item context into compact embeddings and a Recommendation LLM to process these embeddings efficiently. The model replaces beam search decoding with a lightweight item projection head, enabling faster inference while preserving performance. In our implementation, the Item LLM employs a fully-connected layer, and all sub-token embeddings of each item are used as input to the Recommendation LLM.
- **LLMEmb** [18] incorporates collaborative signals into LLM-generated embeddings by modifying the contrastive loss, improving the model's suitability for recommendation tasks. In our implementation, due to computational constraints, we omitted the PCA transformation and the CF-Rec alignment step.
- **HLLM** [2] introduces a two-tier architecture leveraging Large Language Models (LLMs) for sequential recommendation by combining an Item LLM to extract rich content features and a User LLM to model user interests. In our implementation, a fully connected layer is applied within the Item LLM, and the complete set of sub-token embeddings for each item is passed to the User LLM as input.

B Implementation Details

CGRec [23][4], SyNCRec [24][5], UniSRec [6][6], and E4SRec [12][7] were executed with their official code. RecFormer [11], Lite-LLMRec [30], SAID [8], LLMEmb [18], ReLLa [15], A-LLMRec [10], HLLM [2] and CALRec [13] were implemented from scratch. All other CDSR and SR models were implemented using the RecBole framework [34, 35]. Most ID-based CDSR models, with the exception of CGRec and SyNCRec, were specifically designed to capture **only pairwise interactions between two domains.** As a result, these baselines were trained using data from two out of five domains. In these models, each domain was paired with the remaining four domains, and the average performance across these pairings is shown in Table 2 and 3.

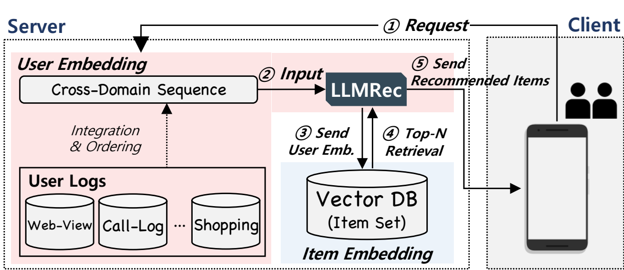

Figure 9: Overview of the inference process

[4] https://github.com/cpark88/CGRec
[5] https://github.com/cpark88/SyNCRec
[6] https://github.com/RUCAIBox/UniSRec
[7] https://github.com/HestiaSky/E4SRec

C The inference and deployment process

In this section, we describe the inference process of our proposed model in Fig. 9. ① The process begins with the client (user side) sending a request to the server, including variables such as `user_id`. On the server side, ② the user's cross-domain sequence data is fed into our model to generate a user embedding. ③ Specifically, the user embedding is derived from the last embedding in the model's output sequence (Section 4.6). ④ The user embedding is then compared to a pre-computed set of item embeddings computed with the LLM backbone (Eq. 11) and stored in a vector database. By performing a similarity calculation, the recommender system identifies the top-N most relevant items for recommendation. ⑤ These recommended items are subsequently sent back to the client, providing personalized recommendation results to the user.

D Details of Pairwise CDSR model

Existing CDSR models predominantly focus on learning from only two domains at a time. In Table 2 and 3, we evaluate the performance of these models on two-domain pairs and report their averaged performance across all pairs as the overall performance for each domain. Due to space limitations, the performance of individual domain pairs was not included in the main body of the paper (Tables 2 and 3). Therefore, this section provides a detailed report of the performance for each domain pair, including the following CDR models: BiGCF [17], DTCDR [37], CMF [27], and DeepAFP [31], as well as the following CDSR models: C^2DSR [1], MAN [14], π-Net [21], and MIFN [20], which were not included in Tables 2 and 3. In the Amazon dataset used for CDR baselines, using the *Games* domain as the source domain generally improved recommendation performance in the *Books* domain when it was the target domain. A similar pattern was observed when the target domains were *Games* and *Outdoors*, with *Toys* and *Games* serving as the respective source domains. In the Telecom dataset, the target domain's recommendation performance was maximized across most baseline models when the combinations of target and source domains were $Web\text{-}View$ and $Benefits$ or $Benefits$ and *Shopping*. The results for CDSR baselines are as follows: In the Amazon dataset, the combinations of target and source domains—*Books* and $Jewelry$, as well as *Outdoors* and $Jewelry$—consistently yielded high recommendation performance across multiple baseline models. Similarly, in the Telecom dataset, the combinations of $Web\text{-}View$ and *Call-Log*, as well as $Benefits$ and $Web\text{-}View$, demonstrated strong performance for the target domain across various models. This detailed analysis highlights the distinct relationships between the domains and provides a deeper insight into the cross-domain interactions.

E Extended Sensitivity Analysis Across All Domains

We conducted a sensitivity analysis on the hyperparameter p, which controls the proportion of cross-domain negative samples during contrastive learning. For completeness, we report the results for all domains (Amazon), including those omitted in the main script, as shown in Fig. 10. As discussed in the main script, we observe consistent performance improvements with increasing p across most domains, although *Call-Log* and *Shopping* domains did not

Table 7: Performance comparison of CDR baselines (BiGCF, DTCDR, CMF, DeepAFP) for two paired domains.

Model			BiGCF						DTCDR						CMF						DeepAFP					
Dataset	Target Domain	Source Domain	HR			NDCG		MRR	HR			NDCG		MRR	HR			NDCG		MRR	HR			NDCG		MRR
			@1	@5	@10	@5	@10	@10	@1	@5	@10	@5	@10	@10	@1	@5	@10	@5	@10	@10	@1	@5	@10	@5	@10	@10
Amazon	Books	Games	0.152	0.332	0.424	0.245	0.275	0.229	0.145	0.288	0.368	0.219	0.245	0.207	0.165	0.328	0.411	0.250	0.277	0.235	0.146	0.290	0.365	0.221	0.245	0.208
		Jewelry	0.124	0.266	0.351	0.198	0.225	0.186	0.144	0.288	0.369	0.219	0.245	0.207	0.162	0.323	0.402	0.246	0.272	0.231	0.142	0.282	0.351	0.215	0.237	0.202
		Outdoors	0.155	0.322	0.414	0.242	0.271	0.227	0.143	0.288	0.367	0.219	0.244	0.206	0.161	0.315	0.391	0.242	0.266	0.227	0.134	0.267	0.336	0.203	0.226	0.191
		Toys	0.160	0.329	0.419	0.248	0.277	0.233	0.144	0.293	0.371	0.222	0.247	0.209	0.160	0.308	0.386	0.237	0.262	0.224	0.140	0.276	0.346	0.211	0.234	0.199
	Games	Books	0.173	0.384	0.512	0.281	0.323	0.264	0.151	0.397	0.537	0.277	0.322	0.256	0.142	0.316	0.429	0.231	0.267	0.218	0.144	0.317	0.427	0.233	0.268	0.220
		Jewelry	0.166	0.367	0.492	0.268	0.309	0.253	0.149	0.394	0.539	0.275	0.321	0.255	0.139	0.314	0.430	0.228	0.266	0.216	0.151	0.328	0.438	0.242	0.277	0.228
		Outdoors	0.168	0.373	0.498	0.273	0.313	0.257	0.150	0.394	0.536	0.275	0.321	0.255	0.141	0.319	0.436	0.231	0.269	0.218	0.147	0.321	0.432	0.236	0.272	0.223
		Toys	0.176	0.388	0.511	0.285	0.325	0.267	0.149	0.393	0.538	0.274	0.321	0.254	0.147	0.329	0.443	0.241	0.277	0.227	0.153	0.333	0.443	0.245	0.281	0.231
	Jewelry	Books	0.097	0.228	0.316	0.164	0.192	0.155	0.098	0.262	0.363	0.182	0.214	0.169	0.113	0.260	0.347	0.189	0.217	0.177	0.108	0.252	0.337	0.182	0.210	0.170
		Games	0.105	0.256	0.348	0.182	0.212	0.170	0.097	0.262	0.362	0.182	0.214	0.169	0.112	0.270	0.364	0.193	0.224	0.180	0.108	0.259	0.351	0.186	0.215	0.174
		Outdoors	0.100	0.240	0.328	0.172	0.200	0.161	0.096	0.262	0.362	0.181	0.214	0.168	0.115	0.263	0.351	0.191	0.219	0.179	0.115	0.263	0.347	0.191	0.218	0.179
		Toys	0.100	0.238	0.330	0.171	0.200	0.161	0.097	0.263	0.361	0.182	0.214	0.169	0.111	0.259	0.346	0.186	0.215	0.174	0.106	0.250	0.339	0.179	0.208	0.168
	Outdoors	Books	0.102	0.236	0.329	0.170	0.200	0.161	0.112	0.287	0.394	0.202	0.236	0.188	0.103	0.245	0.337	0.175	0.205	0.165	0.096	0.236	0.330	0.167	0.198	0.157
		Games	0.115	0.273	0.371	0.196	0.228	0.184	0.112	0.287	0.392	0.202	0.236	0.188	0.111	0.271	0.368	0.193	0.224	0.180	0.109	0.269	0.363	0.191	0.222	0.178
		Jewelry	0.094	0.225	0.313	0.161	0.189	0.151	0.113	0.285	0.391	0.201	0.235	0.187	0.106	0.245	0.336	0.177	0.206	0.167	0.105	0.245	0.334	0.177	0.206	0.166
		Toys	0.104	0.247	0.342	0.177	0.208	0.167	0.112	0.285	0.393	0.201	0.235	0.187	0.105	0.251	0.346	0.179	0.210	0.168	0.101	0.252	0.346	0.178	0.208	0.166
	Toys	Books	0.109	0.255	0.353	0.183	0.215	0.173	0.110	0.270	0.372	0.192	0.225	0.179	0.099	0.244	0.350	0.172	0.206	0.163	0.092	0.232	0.337	0.162	0.196	0.153
		Games	0.113	0.269	0.373	0.193	0.226	0.182	0.107	0.270	0.376	0.190	0.225	0.178	0.109	0.265	0.368	0.189	0.222	0.177	0.103	0.258	0.359	0.182	0.214	0.170
		Jewelry	0.104	0.245	0.340	0.176	0.207	0.166	0.108	0.270	0.375	0.191	0.225	0.179	0.099	0.248	0.352	0.175	0.208	0.165	0.093	0.236	0.342	0.166	0.200	0.156
		Outdoors	0.105	0.251	0.351	0.179	0.211	0.169	0.108	0.264	0.373	0.187	0.222	0.177	0.100	0.248	0.353	0.175	0.209	0.165	0.090	0.221	0.313	0.156	0.186	0.148
Telecom	Web-View	Call-Log	0.849	0.964	0.976	0.915	0.919	0.900	0.717	0.946	0.975	0.846	0.855	0.816	0.794	0.964	0.980	0.892	0.897	0.870	0.839	0.956	0.969	0.908	0.912	0.893
		PoI	0.845	0.964	0.976	0.915	0.919	0.899	0.717	0.946	0.976	0.846	0.856	0.816	0.818	0.965	0.980	0.903	0.908	0.884	0.844	0.956	0.970	0.910	0.914	0.896
		Benefits	0.845	0.965	0.977	0.915	0.919	0.899	0.724	0.947	0.976	0.850	0.860	0.821	0.844	0.965	0.977	0.914	0.918	0.898	0.848	0.958	0.971	0.912	0.917	0.898
		Shopping	0.847	0.964	0.976	0.915	0.919	0.900	0.717	0.946	0.975	0.846	0.855	0.816	0.830	0.965	0.978	0.909	0.913	0.891	0.850	0.958	0.971	0.913	0.918	0.900
	Call-Log	Web-View	0.334	0.686	0.815	0.520	0.562	0.482	0.334	0.674	0.802	0.514	0.555	0.478	0.420	0.753	0.850	0.599	0.630	0.561	0.446	0.701	0.797	0.584	0.615	0.558
		PoI	0.328	0.677	0.809	0.512	0.555	0.475	0.333	0.679	0.813	0.516	0.559	0.480	0.407	0.743	0.846	0.587	0.620	0.549	0.438	0.715	0.809	0.587	0.618	0.557
		Benefits	0.321	0.670	0.805	0.505	0.549	0.468	0.333	0.675	0.804	0.514	0.556	0.478	0.407	0.743	0.841	0.587	0.619	0.548	0.428	0.715	0.817	0.583	0.616	0.552
		Shopping	0.322	0.669	0.806	0.505	0.549	0.469	0.325	0.669	0.804	0.506	0.550	0.470	0.409	0.738	0.838	0.586	0.618	0.548	0.465	0.733	0.822	0.609	0.638	0.580
	PoI	Web-View	0.333	0.649	0.777	0.500	0.541	0.467	0.341	0.651	0.786	0.505	0.548	0.474	0.378	0.632	0.765	0.510	0.553	0.487	0.479	0.705	0.801	0.601	0.632	0.579
		Call-Log	0.321	0.635	0.773	0.485	0.530	0.454	0.351	0.669	0.801	0.518	0.561	0.486	0.437	0.733	0.827	0.551	0.627	0.564	0.466	0.701	0.797	0.593	0.624	0.570
		Benefits	0.326	0.652	0.775	0.491	0.535	0.460	0.364	0.682	0.803	0.533	0.572	0.500	0.443	0.726	0.822	0.596	0.627	0.565	0.471	0.704	0.797	0.596	0.626	0.572
		Shopping	0.323	0.633	0.771	0.485	0.530	0.455	0.337	0.645	0.783	0.499	0.544	0.469	0.431	0.720	0.817	0.587	0.618	0.555	0.476	0.702	0.792	0.599	0.628	0.628
	Benefits	Web-View	0.527	0.856	0.925	0.706	0.728	0.665	0.368	0.839	0.932	0.625	0.656	0.566	0.661	0.859	0.933	0.764	0.788	0.743	0.654	0.851	0.921	0.756	0.779	0.734
		Call-Log	0.577	0.853	0.915	0.726	0.747	0.693	0.405	0.849	0.938	0.646	0.675	0.590	0.667	0.864	0.931	0.771	0.793	0.749	0.649	0.827	0.899	0.741	0.765	0.723
		PoI	0.580	0.862	0.921	0.734	0.753	0.699	0.369	0.833	0.933	0.623	0.656	0.566	0.651	0.859	0.927	0.759	0.781	0.735	0.660	0.806	0.869	0.734	0.755	0.720
		Shopping	0.602	0.845	0.899	0.733	0.751	0.704	0.395	0.830	0.927	0.631	0.663	0.578	0.658	0.850	0.914	0.758	0.779	0.736	0.663	0.828	0.880	0.748	0.765	0.729
	Shopping	Web-View	0.185	0.394	0.527	0.292	0.335	0.276	0.121	0.335	0.480	0.229	0.276	0.214	0.185	0.347	0.488	0.266	0.312	0.258	0.181	0.339	0.481	0.260	0.306	0.253
		Call-Log	0.201	0.383	0.503	0.294	0.332	0.280	0.118	0.330	0.478	0.225	0.273	0.211	0.170	0.322	0.472	0.245	0.293	0.239	0.182	0.319	0.450	0.250	0.292	0.244
		PoI	0.204	0.373	0.486	0.290	0.326	0.277	0.116	0.339	0.481	0.229	0.275	0.211	0.176	0.357	0.499	0.266	0.312	0.256	0.182	0.333	0.472	0.257	0.302	0.251
		Benefits	0.202	0.372	0.486	0.289	0.326	0.277	0.117	0.343	0.491	0.231	0.279	0.215	0.170	0.363	0.499	0.267	0.311	0.254	0.187	0.353	0.488	0.270	0.314	0.261

exhibit a clear upward trend, whereas the others showed a robust increase pattern.

F Extended Analysis on Cold & Warm Scenario Across All Domains

Cold/Warm Item Scenario While Section 5.6 presents the evaluation results on two representative domains due to space constraints, we report here the results across all domains under the cold- and warm-item scenarios. As shown in Fig. 11, the proposed model demonstrates its ability to capture item-level collaborative signals, achieving superior performance over IDRec models for both cold and warm items.

Cold User Scenario To complement the findings in Section 5.6, which are limited to two domains due to space constraints, we present comprehensive results across all domains, as shown in Fig. 12. DuELRec maintains consistent superiority over IDRec, with particularly substantial gains in domains where cold-user performance is hindered by limited collaborative signals.

G Extended Analysis on Instruction-Tuning Across All Domains

In Section 4.3, we proposed an auxiliary task that integrates collaborative signals with recommendation instructions to enable LLMRec to better adapt to textual user behavior sequences. This task involves fine-tuning the LLM using PEFT methods such as LoRA and prompt-tuning. We conducted experiments to compare performance with and without this auxiliary task. As shown in Fig. 13, the auxiliary task with PEFT significantly improved performance in all domains. This shows that allowing the LLM to learn from user behavior logs effectively improves recommendation performance.

H Future Work

In LLMRec, there are two primary approaches: (1) retrieval-based and (2) generative. The retrieval-based approach operates by recommending items strictly from a predefined set, ensuring that all recommendations are confined to this set. This makes it particularly advantageous when dealing with a large-scale item pool, as it eliminates the risk of hallucination. All the baselines introduced in Section 5 (Experiments), including our proposed model, adopt this retrieval-based strategy.

On the other hand, the generative approach involves the LLM generating textual titles (or descriptions) of recommended items, typically selected from a relatively small candidate set (around 100 items). In this method, the candidate items are explicitly provided to the LLM as part of the prompt, along with instructions to identify the most suitable item from the given candidates. This approach holds significant promise for building personalized recommendation agents. However, it comes with a potential drawback: the risk

Table 8: Performance comparison of CDSR baselines (C^{2}DSR, MAN, π-Net, MIFN) for two paired domains.

Model			C^{2}DSR						MAN						π-Net						MIFN					
Dataset	Target Domain	Source Domain	HR @1	HR @5	HR @10	NDCG @5	NDCG @10	MRR @10	HR @1	HR @5	HR @10	NDCG @5	NDCG @10	MRR @10	HR @1	HR @5	HR @10	NDCG @5	NDCG @10	MRR @10	HR @1	HR @5	HR @10	NDCG @5	NDCG @10	MRR @10
Amazon	Books	Games	0.115	0.292	0.395	0.212	0.238	0.206	0.112	0.254	0.315	0.233	0.253	0.229	0.093	0.203	0.279	0.149	0.174	0.142	0.092	0.199	0.277	0.147	0.172	0.140
		Jewelry	0.176	0.344	0.434	0.262	0.291	0.264	0.120	0.287	0.361	0.252	0.306	0.274	0.128	0.266	0.351	0.200	0.227	0.189	0.112	0.258	0.348	0.187	0.216	0.176
		Outdoors	0.144	0.303	0.414	0.225	0.261	0.233	0.115	0.354	0.400	0.234	0.259	0.249	0.114	0.235	0.322	0.177	0.205	0.169	0.074	0.178	0.222	0.128	0.142	0.117
		Toys	0.180	0.340	0.436	0.263	0.294	0.267	0.103	0.367	0.421	0.259	0.276	0.209	0.110	0.239	0.322	0.177	0.203	0.167	0.108	0.239	0.325	0.176	0.204	0.167
	Games	Books	0.131	0.359	0.494	0.248	0.292	0.250	0.122	0.358	0.407	0.243	0.258	0.267	0.141	0.379	0.505	0.263	0.303	0.241	0.164	0.404	0.535	0.287	0.329	0.266
		Jewelry	0.107	0.321	0.465	0.216	0.262	0.222	0.128	0.365	0.412	0.250	0.265	0.273	0.151	0.407	0.548	0.283	0.329	0.261	0.150	0.403	0.543	0.281	0.326	0.260
		Outdoors	0.148	0.390	0.527	0.271	0.315	0.269	0.129	0.364	0.509	0.253	0.367	0.277	0.147	0.393	0.530	0.275	0.319	0.254	0.160	0.408	0.541	0.287	0.331	0.266
		Toys	0.137	0.376	0.518	0.260	0.306	0.260	0.116	0.364	0.519	0.239	0.357	0.263	0.135	0.381	0.511	0.262	0.304	0.240	0.138	0.377	0.507	0.261	0.303	0.240
	Jewelry	Books	0.096	0.243	0.363	0.169	0.208	0.180	0.118	0.258	0.300	0.138	0.251	0.220	0.085	0.231	0.334	0.160	0.193	0.150	0.088	0.229	0.331	0.160	0.193	0.151
		Games	0.080	0.213	0.326	0.146	0.183	0.159	0.051	0.256	0.297	0.163	0.176	0.192	0.073	0.188	0.276	0.132	0.160	0.125	0.095	0.209	0.294	0.154	0.181	0.146
		Outdoors	0.097	0.245	0.355	0.172	0.208	0.183	0.096	0.256	0.311	0.135	0.254	0.183	0.089	0.234	0.335	0.163	0.196	0.153	0.104	0.245	0.339	0.175	0.206	0.165
		Toys	0.088	0.234	0.336	0.194	0.225	0.173	0.052	0.272	0.307	0.116	0.228	0.142	0.090	0.226	0.324	0.160	0.191	0.151	0.098	0.235	0.330	0.168	0.198	0.158
	Outdoors	Books	0.103	0.256	0.373	0.182	0.219	0.192	0.112	0.248	0.390	0.130	0.244	0.154	0.099	0.257	0.352	0.180	0.211	0.211	0.111	0.259	0.353	0.187	0.217	0.176
		Games	0.108	0.268	0.380	0.191	0.227	0.199	0.110	0.240	0.378	0.151	0.237	0.153	0.079	0.206	0.286	0.144	0.169	0.134	0.102	0.223	0.298	0.164	0.188	0.155
		Jewelry	0.109	0.245	0.352	0.178	0.212	0.191	0.117	0.258	0.295	0.138	0.250	0.159	0.110	0.280	0.378	0.197	0.229	0.183	0.106	0.272	0.371	0.191	0.223	0.178
		Toys	0.100	0.253	0.362	0.178	0.213	0.190	0.114	0.206	0.305	0.134	0.250	0.159	0.090	0.239	0.333	0.166	0.197	0.155	0.110	0.258	0.347	0.186	0.214	0.174
	Toys	Books	0.089	0.254	0.373	0.172	0.211	0.182	0.138	0.268	0.314	0.177	0.292	0.202	0.077	0.220	0.327	0.150	0.184	0.141	0.102	0.246	0.347	0.175	0.208	0.165
		Games	0.109	0.281	0.389	0.197	0.232	0.203	0.102	0.260	0.301	0.139	0.252	0.266	0.080	0.203	0.287	0.143	0.170	0.134	0.086	0.206	0.295	0.147	0.176	0.140
		Jewelry	0.108	0.267	0.380	0.189	0.225	0.200	0.149	0.273	0.310	0.198	0.209	0.152	0.101	0.265	0.364	0.185	0.217	0.172	0.084	0.241	0.347	0.164	0.198	0.153
		Outdoors	0.097	0.257	0.372	0.179	0.216	0.190	0.100	0.266	0.310	0.168	0.283	0.193	0.099	0.255	0.346	0.179	0.209	0.167	0.095	0.247	0.342	0.173	0.204	0.161
Telecom	Web-View	Call-Log	0.786	0.947	0.971	0.877	0.885	0.858	0.781	0.924	0.969	0.835	0.880	0.847	0.636	0.896	0.946	0.780	0.796	0.747	0.641	0.898	0.948	0.783	0.799	0.751
		PoI	0.776	0.942	0.968	0.869	0.878	0.850	0.736	0.915	0.975	0.846	0.860	0.844	0.623	0.888	0.943	0.769	0.787	0.737	0.624	0.889	0.942	0.770	0.788	0.737
		Benefits	0.738	0.923	0.956	0.841	0.852	0.714	0.755	0.925	0.968	0.855	0.863	0.706	0.603	0.879	0.937	0.755	0.774	0.722	0.600	0.884	0.939	0.756	0.774	0.721
		Shopping	0.754	0.934	0.961	0.853	0.862	0.832	0.751	0.924	0.958	0.850	0.862	0.846	0.592	0.874	0.930	0.746	0.765	0.711	0.586	0.872	0.933	0.744	0.764	0.709
	Call-Log	Web-View	0.466	0.693	0.796	0.585	0.618	0.573	0.382	0.697	0.782	0.575	0.612	0.512	0.451	0.685	0.785	0.572	0.605	0.548	0.553	0.779	0.863	0.675	0.703	0.652
		PoI	0.453	0.700	0.808	0.583	0.618	0.567	0.395	0.710	0.793	0.538	0.566	0.515	0.456	0.679	0.782	0.572	0.606	0.551	0.431	0.676	0.784	0.559	0.594	0.535
		Benefits	0.355	0.621	0.752	0.494	0.536	0.481	0.375	0.631	0.736	0.492	0.507	0.479	0.423	0.653	0.765	0.543	0.579	0.522	0.425	0.655	0.767	0.545	0.582	0.524
		Shopping	0.480	0.715	0.805	0.603	0.632	0.588	0.474	0.693	0.797	0.525	0.575	0.534	0.413	0.644	0.757	0.534	0.570	0.512	0.415	0.643	0.753	0.534	0.569	0.512
	PoI	Web-View	0.356	0.625	0.751	0.498	0.538	0.484	0.341	0.607	0.797	0.482	0.564	0.556	0.439	0.702	0.806	0.578	0.612	0.551	0.423	0.689	0.797	0.565	0.600	0.538
		Call-Log	0.427	0.690	0.798	0.567	0.602	0.550	0.413	0.692	0.750	0.504	0.595	0.549	0.414	0.676	0.786	0.553	0.588	0.526	0.385	0.654	0.769	0.527	0.564	0.500
		Benefits	0.436	0.702	0.801	0.558	0.610	0.560	0.391	0.653	0.754	0.504	0.613	0.543	0.399	0.662	0.773	0.538	0.574	0.512	0.401	0.659	0.770	0.538	0.574	0.512
		Shopping	0.429	0.689	0.785	0.567	0.598	0.550	0.418	0.705	0.755	0.508	0.593	0.548	0.380	0.645	0.761	0.520	0.557	0.493	0.386	0.643	0.756	0.521	0.558	0.496
	Benefits	Web-View	0.518	0.768	0.881	0.644	0.681	0.626	0.607	0.764	0.881	0.632	0.652	0.604	0.552	0.753	0.850	0.654	0.685	0.635	0.549	0.758	0.851	0.654	0.685	0.633
		Call-Log	0.571	0.813	0.899	0.694	0.722	0.672	0.532	0.824	0.872	0.708	0.732	0.687	0.522	0.756	0.867	0.640	0.676	0.617	0.520	0.764	0.875	0.643	0.679	0.618
		PoI	0.591	0.821	0.903	0.709	0.735	0.688	0.587	0.809	0.907	0.685	0.741	0.681	0.533	0.771	0.875	0.654	0.688	0.630	0.544	0.771	0.877	0.659	0.694	0.637
		Shopping	0.570	0.811	0.898	0.694	0.722	0.673	0.550	0.814	0.906	0.641	0.696	0.671	0.471	0.729	0.847	0.601	0.640	0.575	0.469	0.752	0.876	0.611	0.652	0.582
	Shopping	Web-View	0.162	0.383	0.538	0.280	0.330	0.286	0.139	0.295	0.465	0.270	0.302	0.261	0.206	0.456	0.597	0.335	0.380	0.313	0.237	0.485	0.628	0.367	0.414	0.347
		Call-Log	0.186	0.425	0.573	0.308	0.356	0.307	0.169	0.399	0.547	0.266	0.338	0.291	0.204	0.429	0.559	0.320	0.362	0.301	0.210	0.447	0.579	0.332	0.375	0.312
		PoI	0.148	0.390	0.545	0.272	0.322	0.272	0.164	0.307	0.545	0.271	0.317	0.291	0.195	0.426	0.572	0.314	0.361	0.296	0.174	0.317	0.657	0.247	0.353	0.266
		Benefits	0.228	0.484	0.630	0.362	0.409	0.358	0.226	0.440	0.556	0.310	0.391	0.341	0.189	0.430	0.573	0.312	0.358	0.292	0.186	0.421	0.569	0.306	0.353	0.287

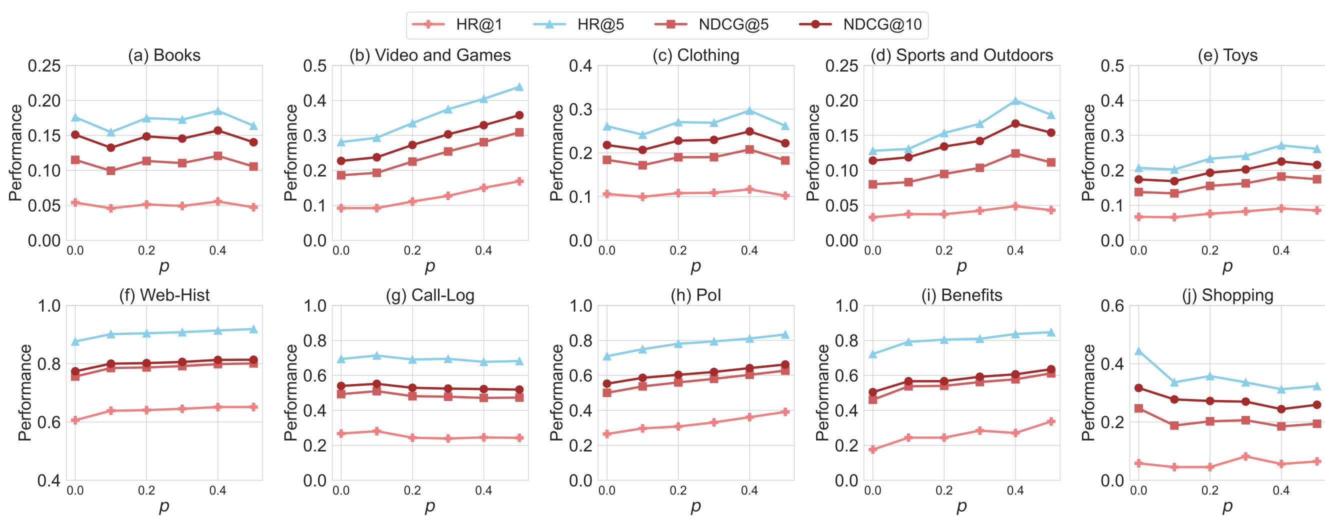

Figure 10: Domain-Wise Performance Sensitivity analysis of p for negative sampling

of hallucination, where the model may generate recommendations for items that are not part of the predefined candidate set.

To evaluate the performance of generative LLMRec, we conducted experiments using a representative generative model, **A-LLMRec** [10]. A-LLMRec introduces a two-stage framework that aligns collaborative filtering models with LLMs, leveraging an alignment network to transfer collaborative knowledge into the token space of LLMs without requiring fine-tuning of either component. During the validation phase, this model generates the title of the *top*-1 item among the 100 candidates included in the prompt. Note

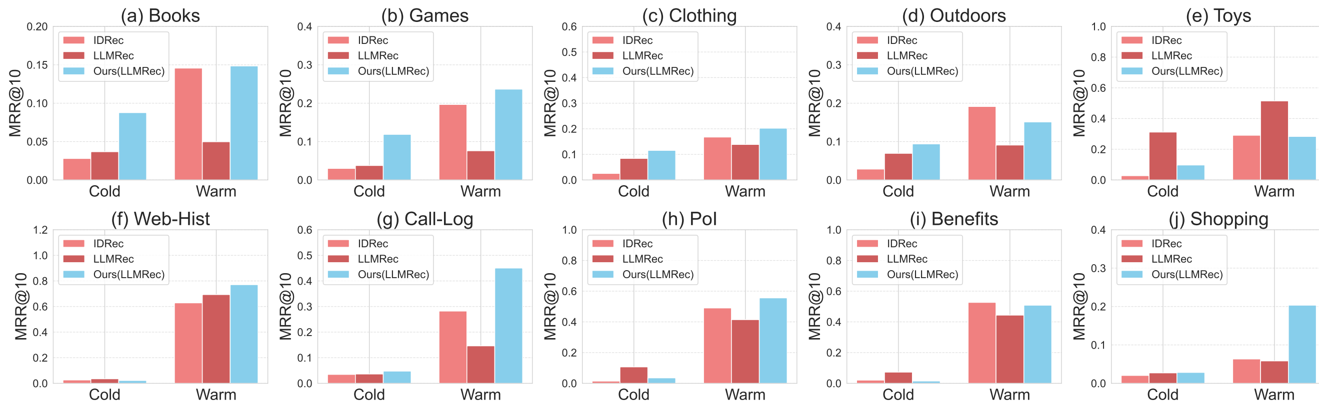

Figure 11: Domain-Wise Performance Comparison under Cold/Warm-Item Scenario

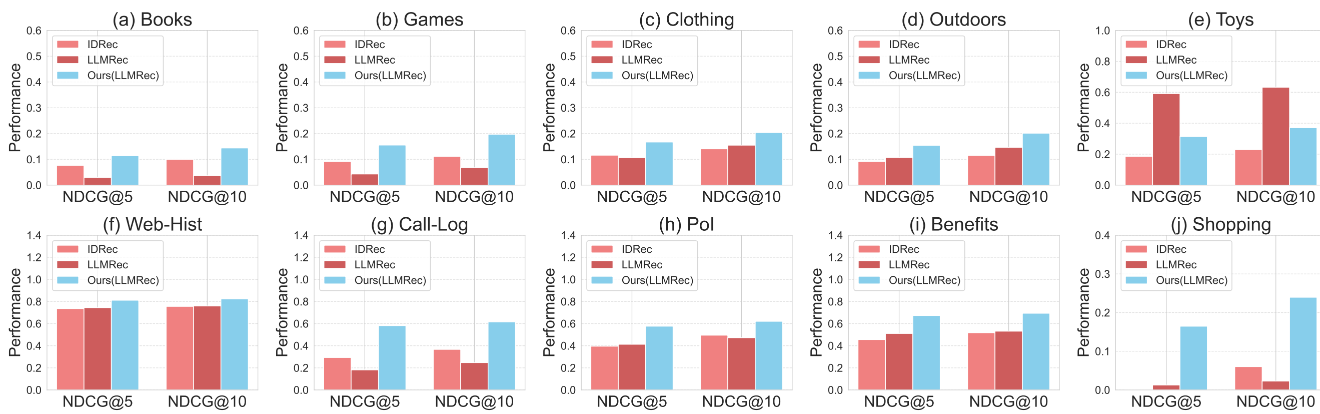

Figure 12: Domain-Wise Performance Comparison under Cold-User Scenario

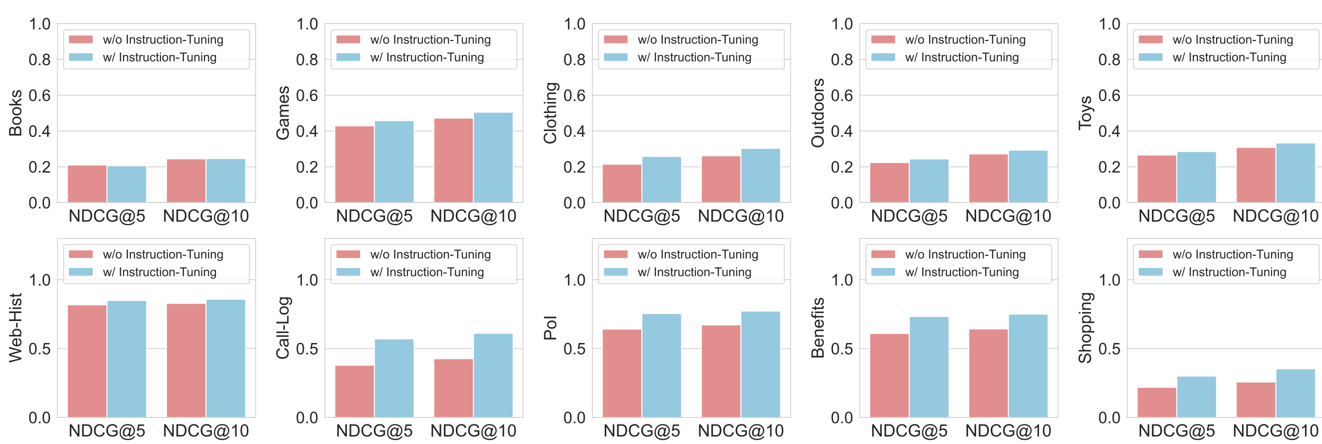

Figure 13: Domain-Wise Performance Comparison: Without vs. With Instruction-Tuning

that the candidate set in the prompt is composed of one positive sample and 99 negative samples, following the same experimental setting used during the validation phase in our study.

Table 9 presents a comparison of the HR@1 performance between A-LLMRec and our proposed model. While its performance was lower compared to our retrieval-based model, it demonstrated reasonable effectiveness, even with the inherent risk of hallucination due to direct item generation. These findings highlight the potential of generative LLMRec for future research. Our next goal is to develop generative methods capable of recommending items from extensive item pools without hallucination. We believe that such advancements in generative LLMRec could play a crucial

Table 9: Comparison of the HR@1 performance between retrieval-based (Ours) and generative model (A-LLMRec).

Model		Retrieval -Based	Generative	Gap(%)
Dataset	**Domain**			
Amazon	**Books**	0.1096	-	-
	Games	0.2941	0.0730	-302.88
	Jewelry	0.1487	0.0762	-95.14
	Outdoors	0.1223	0.0639	-91.39
	Toys	0.1624	0.1138	-42.71
Telecom	**Web-View**	0.7219	0.6840	-5.54
	Call-Log	0.3301	0.1800	-83.59
	PoI	0.5685	0.2000	-184.82
	Benefits	0.4653	-	-
	Shopping	0.1632	0.0270	-504.44

* The - symbol indicates cases where performance evaluation was not performed due to our limited computing resources, or where performance was too low due to problems such as the hallucination problem in Generative LLMRec.

role in creating highly personalized and reliable recommendation agents.